ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ 3 SET C **SUMMER SEMESTER, 2023-2024**

DURATION: 30 MINUTES **FULL MARKS: 20**

CSE 4205: Digital Logic Design

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID:

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1. You have to design a $2^3 \times 3$ priority encoder with a Valid output. The encoder has 8 inputs labeled I_0 to I_7, 3 output lines labeled O_2, O_1, O_0, and a Valid signal V. 4.5 (CO5) (PO3)

   The inputs have the following randomly assigned priority (from highest to lowest):

   $$\text{Priority: } I_7 > I_4 > I_1 > I_0 > I_3 > I_2 > I_5 > I_6$$

   To be noted -

   1. If multiple inputs are **HIGH**, the encoder outputs the binary code of the **highest-priority input**.
   2. The **Valid** output V is 1 if any **input** is **HIGH**, and **0** if **all are LOW**. You can arrange the inputs however you like.

   Fill in the truth table based on this priority.

Inputs								Outputs			
I_7	I_4	I_1	I_0	I_3	I_2	I_5	I_6	O_2	O_1	O_0	V
0	0	0	0	0	0	0	0	X	X	X	0
1	X	X	X	X	X	X	X	1	1	1	1
0	1	X	X	X	X	X	X	1	0	0	1
0	0	1	X	X	X	X	X	0	0	1	1
0	0	0	1	X	X	X	X	0	0	0	1
0	0	0	0	1	X	X	X	0	1	1	1
0	0	0	0	0	1	X	X	0	1	0	1
0	0	0	0	0	0	1	X	1	0	1	1
0	0	0	0	0	0	0	1	1	1	0	1

Solution:

Rubrics:

- 0.3 point (times 9) for each correct input row.
- 0.2 point (times 9) for each correct output row,

2. Design a combinational circuit using comparator ICs and logic gates to determine the **largest among three input numbers**: A, B, and C. The circuit must satisfy the following specifications: 7 + 1 + 2 (CO5) (PO3)

- **Inputs:** Three binary numbers A, B, and C (you are not required to consider the number of bits, assume they are compatible with the comparator IC).
- **Outputs:** Three output signals:
  - **A largest**: High (1) if $A > B$ and $A > C$
  - **B largest**: High (1) if $B > A$ and $B > C$
  - **C largest**: High (1) if $C > A$ and $C > B$
- It is guaranteed that no two inputs are equal. Hence, exactly one of the outputs will be high at a time.
- **Comparator IC:** You may only use 2-input comparator blocks. Each comparator has two n-bit number inputs (X and Y) and three outputs: $X > Y$, $X < Y$, and $X = Y$, where X and Y are two n-bit numbers.

  You are not required to design the comparator IC, it may be represented as a block in your diagram.
- **Design Requirement:** Using these comparator IC blocks and any other component as you wish, draw the **block diagram** of a combinational circuit that produces the required outputs. You may assume ideal behavior and unlimited availability of logic gates and comparators.

Note: A block diagram means that you do not need to draw the internal structure of the comparator ICs. Instead, use labeled blocks showing inputs and outputs clearly.

Answer the following questions -

a) Design the above-mentioned circuit.

b) Explicitly mention the number of 2-number comparator ICs you have used.

c) Briefly explain the working mechanism (avoid unnecessary details).

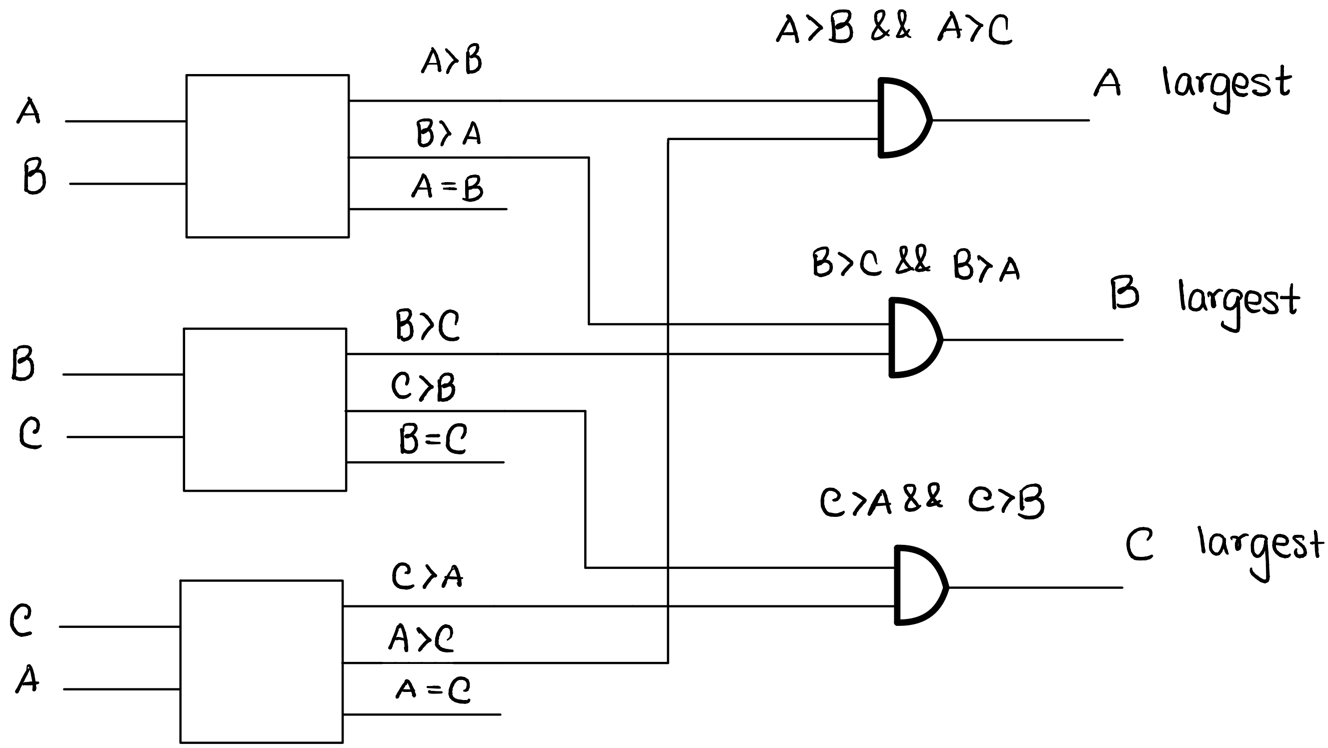
d) **Bonus 5 Marks** for proposing the most efficient solution.

Solution:

Rubrics:

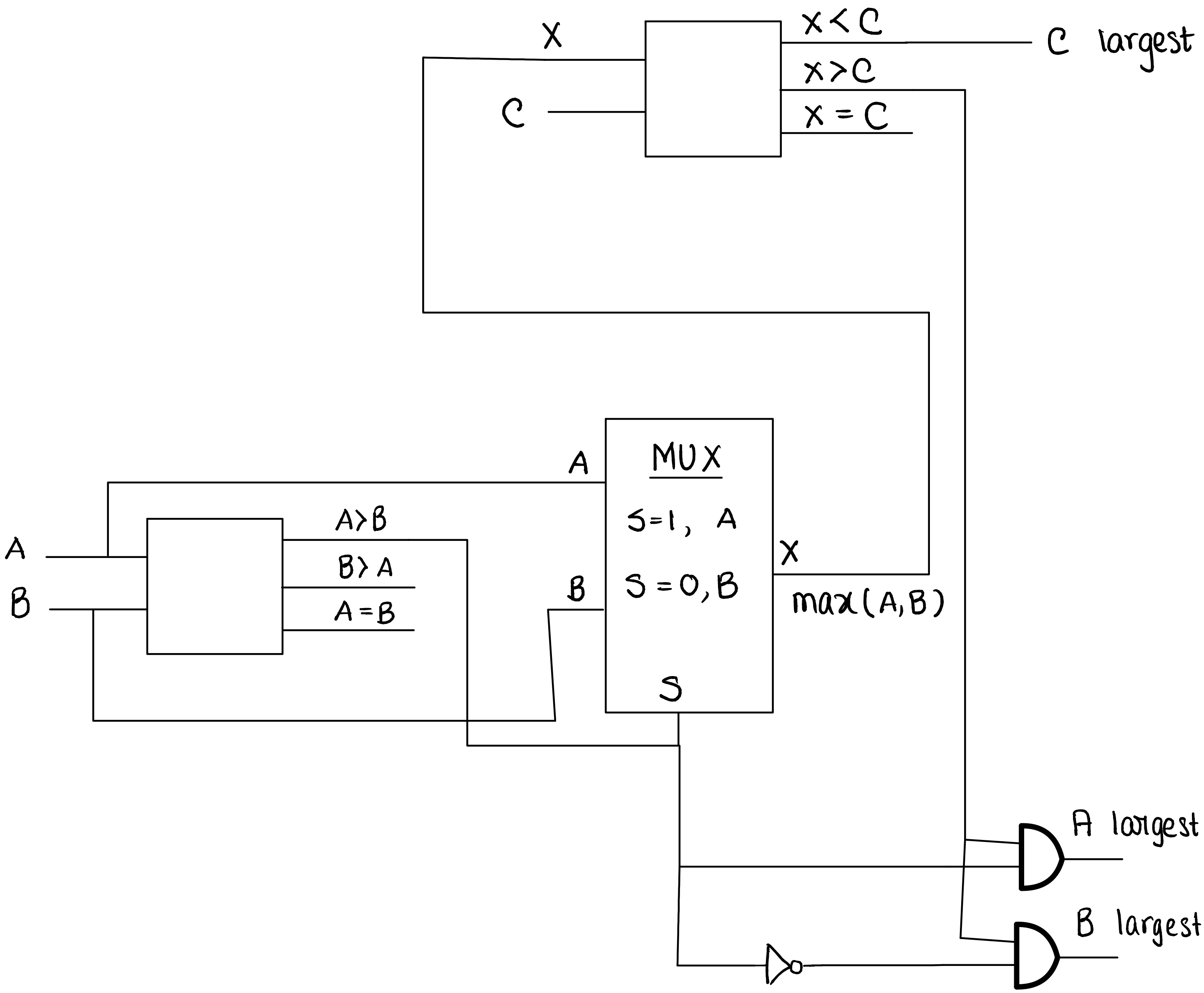
- 2 points (times 3) for comparison between each pair of input.
- 1 points for correctly determining the output for each output.
- 1 points for correctly mentioning the correct number of comparators used.
- 2 points for accurate logical justification.
- **5 bonus points** for efficient solution using only 2 comparators.

Naive Approach



Comparators used: 3

Efficient Solution



Comparators used : 2

3. Given, 3.5 + 2

(CO5)
(PO3)

$$F_1(A,B,C) = \sum m(4,6,7)$$
$$F_2(A,B,C) = \sum m(1,5,6)$$

Complete the PLD circuit for F_1 and F_2 by making the necessary connections (using 'X' at junctions) using the following **PLA (Programmable Logic Array)** structure.

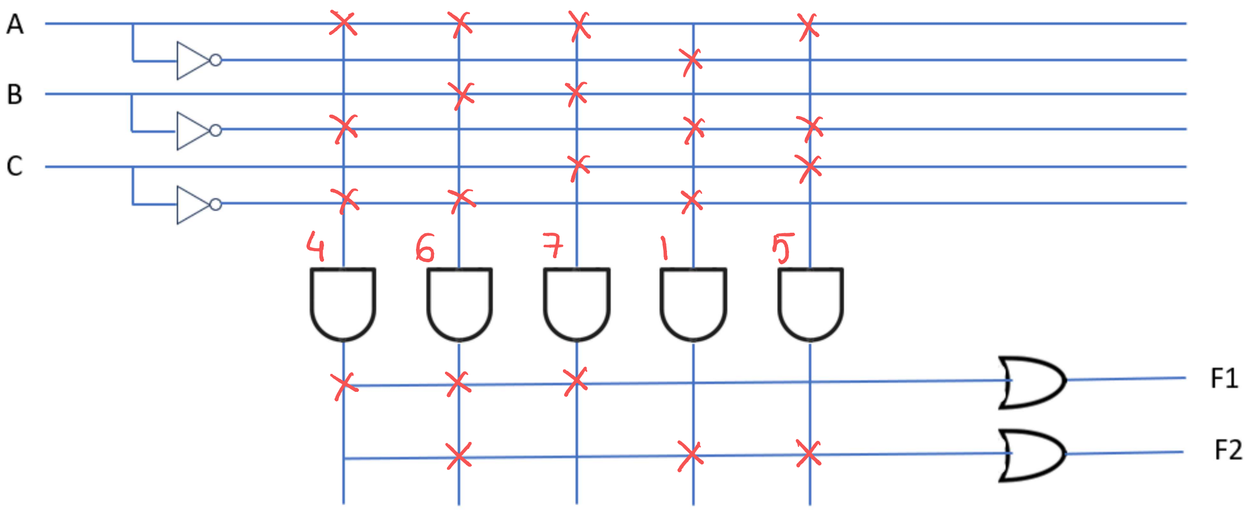


Figure 1: PLA Circuit

Solution:

Rubrics:

- 0.5 point (times 5) for each correct AND array combination.
- 0.75 point (times 2) for each correct OR array output.
- 0.5 point each (times 3) for correctly defining n, m and p and stating their values.
- 0.25 point (times 2) for each correct MCQ answer.

And answer the following questions-

a) A PLA circuit structure is defined by $n \times p \times m$. Write down what each symbol represents and provide their values according to the given PLA circuit diagram that you have constructed.

- n: Input, **Value:** 3
- p: Intermediate Output of AND array, **Value:** 5
- m: Output of OR array, **Value:** 2

b) Put a tick next to the correct option.
For a **PAL (Programmable Array Logic)**:

- AND Array: ☐ Fixed ☑ Programmable
- OR Array: ☑ Fixed ☐ Programmable